

## BADAJOZ AP, Spain

WMO: 083300

Lat: 38.883N

Lon: 6.814W

Elev: 630

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 29.8       | 32.1 | 21.8                          | 16.7 | 42.2 | 26.0 | 20.4 | 43.5 | 24.1                  | 55.3 | 21.1 | 53.1 | 1.9                   | 100  | 0.435 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 30.3                   | 102.5           | 71.9 | 99.4 | 70.2 | 96.7 | 69.2 | 74.6                | 98.4 | 72.5 | 95.1 | 70.7 | 92.1 | 7.5                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 100.1 | 81.9 | 65.1 | 95.1 | 78.6 | 64.0 | 91.4 | 77.0 | 38.3          | 98.6 | 36.3 | 95.3 | 34.9 | 92.0 | 88.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 19.5          | 16.9 | 14.8 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 63.3   | 47.7  | 50.1  | 55.2  | 60.7  | 67.1  | 75.0  | 79.0  | 79.9  | 74.7  | 65.7  | 55.3  | 48.9  |  |  |
|                                                                        | DBStd       | 12.62  | 4.95  | 4.98  | 4.64  | 5.37  | 6.20  | 6.39  | 5.13  | 5.09  | 4.70  | 5.48  | 5.40  | 5.03  |  |  |
|                                                                        | HDD50       | 275    | 106   | 57    | 12    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 16    | 84    |  |  |
|                                                                        | HDD65       | 2311   | 537   | 418   | 305   | 150   | 50    | 2     | 0     | 0     | 0     | 57    | 292   | 500   |  |  |
|                                                                        | CDD50       | 5145   | 33    | 59    | 172   | 322   | 531   | 751   | 898   | 926   | 742   | 486   | 176   | 49    |  |  |
|                                                                        | CDD65       | 1708   | 0     | 0     | 0     | 22    | 116   | 303   | 433   | 461   | 292   | 79    | 2     | 0     |  |  |
|                                                                        | CDH74       | 21608  | 0     | 0     | 47    | 420   | 1553  | 3849  | 5669  | 6007  | 3254  | 801   | 8     | 0     |  |  |
|                                                                        | CDH80       | 11536  | 0     | 0     | 2     | 104   | 636   | 2072  | 3323  | 3548  | 1580  | 271   | 0     | 0     |  |  |
|                                                                        | WSAvg       | 6.1    | 5.4   | 6.6   | 7.3   | 6.8   | 6.7   | 6.8   | 6.8   | 6.2   | 5.2   | 5.1   | 5.5   | 5.1   |  |  |
|                                                                        | PrecAvg     | 19.2   | 2.1   | 2.2   | 1.9   | 1.9   | 1.3   | 0.9   | 0.2   | 0.2   | 0.9   | 2.5   | 2.5   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecMax     | 31.7   | 8.2   | 6.6   | 7.1   | 5.7   | 4.3   | 5.3   | 1.1   | 1.3   | 4.8   | 10.1  | 8.9   | 9.6   |  |  |
|                                                                        | PrecMin     | 9.0    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   |  |  |
|                                                                        | PrecStd     | 5.6    | 1.6   | 1.7   | 1.7   | 1.3   | 1.1   | 0.9   | 0.3   | 0.3   | 1.0   | 1.9   | 1.8   | 2.1   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 64.6  | 70.0  | 79.1  | 86.4  | 95.4  | 104.3 | 106.2 | 108.0 | 100.5 | 92.4  | 75.1  | 65.1  |  |  |
|                                                                        |             | MCWB   | 54.5  | 55.7  | 58.2  | 62.2  | 67.3  | 73.2  | 74.5  | 75.0  | 70.2  | 66.0  | 61.6  | 57.7  |  |  |
|                                                                        | 2%          | DB     | 61.2  | 66.2  | 75.2  | 83.5  | 91.1  | 98.8  | 102.1 | 103.3 | 96.4  | 87.2  | 71.1  | 62.5  |  |  |
|                                                                        |             | MCWB   | 53.5  | 53.9  | 57.2  | 61.7  | 65.2  | 70.7  | 71.2  | 73.0  | 68.7  | 63.1  | 59.0  | 56.9  |  |  |
|                                                                        | 5%          | DB     | 59.1  | 63.2  | 71.5  | 79.0  | 86.6  | 95.4  | 98.8  | 99.7  | 92.7  | 82.5  | 67.9  | 60.4  |  |  |
|                                                                        |             | MCWB   | 52.8  | 52.9  | 56.1  | 60.5  | 64.1  | 69.2  | 70.4  | 69.9  | 67.5  | 63.3  | 58.5  | 54.7  |  |  |
|                                                                        | 10%         | DB     | 57.1  | 60.6  | 67.7  | 74.8  | 82.8  | 91.7  | 95.4  | 96.4  | 89.3  | 78.4  | 65.1  | 58.3  |  |  |
|                                                                        |             | MCWB   | 51.9  | 52.7  | 55.0  | 59.0  | 62.7  | 67.7  | 69.2  | 69.4  | 66.7  | 61.7  | 57.4  | 53.1  |  |  |
|                                                                        | 0.4%        | WB     | 58.1  | 59.3  | 61.2  | 65.9  | 70.4  | 75.7  | 76.6  | 78.9  | 73.5  | 68.5  | 65.3  | 60.9  |  |  |
|                                                                        |             | MCDB   | 60.4  | 63.5  | 73.2  | 80.2  | 91.9  | 100.1 | 102.5 | 105.3 | 96.5  | 80.1  | 69.8  | 62.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 56.2  | 57.1  | 59.2  | 63.4  | 67.5  | 72.8  | 73.9  | 75.1  | 70.7  | 67.0  | 62.7  | 58.0  |  |  |
|                                                                        |             | MCDB   | 58.9  | 61.8  | 69.7  | 78.6  | 85.6  | 95.6  | 98.3  | 97.6  | 91.2  | 77.6  | 66.8  | 61.3  |  |  |
|                                                                        | 5%          | WB     | 54.7  | 55.3  | 57.6  | 61.4  | 65.6  | 70.7  | 71.9  | 72.4  | 69.0  | 65.6  | 60.6  | 56.0  |  |  |
|                                                                        |             | MCDB   | 57.6  | 60.4  | 67.6  | 75.5  | 82.3  | 92.0  | 94.7  | 94.5  | 87.4  | 75.7  | 65.2  | 59.2  |  |  |
|                                                                        | 10%         | WB     | 53.0  | 53.6  | 56.1  | 59.8  | 63.7  | 68.9  | 69.9  | 70.5  | 67.5  | 64.2  | 58.9  | 54.2  |  |  |
|                                                                        |             | MCDB   | 56.2  | 59.2  | 65.0  | 72.1  | 79.6  | 88.5  | 91.3  | 91.6  | 84.5  | 73.9  | 63.6  | 57.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 18.5  | 20.8  | 21.9  | 23.2  | 25.4  | 28.3  | 31.2  | 30.3  | 27.3  | 22.8  | 19.1  | 18.3  |  |  |
|                                                                        |             | MCDBR  | 21.6  | 25.0  | 30.4  | 31.1  | 32.4  | 33.4  | 34.6  | 34.7  | 32.6  | 31.0  | 22.7  | 19.1  |  |  |
|                                                                        |             | MCWBR  | 14.4  | 14.5  | 15.4  | 13.3  | 12.5  | 11.4  | 12.7  | 12.1  | 12.4  | 12.4  | 12.5  | 12.5  |  |  |
|                                                                        |             | MCDBR  | 14.2  | 18.9  | 24.3  | 26.5  | 29.0  | 32.1  | 33.1  | 31.9  | 28.8  | 22.7  | 15.6  | 14.0  |  |  |
|                                                                        | 5% WB       | MCWBR  | 10.3  | 12.1  | 13.3  | 12.0  | 11.8  | 11.5  | 13.7  | 12.8  | 11.5  | 10.7  | 9.9   | 10.4  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.314 | 0.332 | 0.365 | 0.367 | 0.376 | 0.379 | 0.358 | 0.374 | 0.368 | 0.353 | 0.327 | 0.314 |  |  |
|                                                                        | taud        |        | 2.493 | 2.444 | 2.357 | 2.366 | 2.370 | 2.378 | 2.403 | 2.368 | 2.408 | 2.471 | 2.500 | 2.501 |  |  |
|                                                                        | Ebn at Noon |        | 274   | 283   | 283   | 289   | 287   | 285   | 290   | 283   | 278   | 272   | 266   | 265   |  |  |
|                                                                        | Edh at Noon |        | 25    | 30    | 36    | 38    | 38    | 38    | 37    | 37    | 34    | 29    | 24    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 731   | 1036  | 1387  | 1746  | 2122  | 2416  | 2511  | 2198  | 1683  | 1153  | 809   | 649   |  |  |
|                                                                        | RadStd      |        | 102   | 133   | 159   | 163   | 178   | 123   | 63    | 70    | 107   | 102   | 102   | 76    |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page